Designing Microsoft Azure Infrastructure Solutions: AZ:305

TECHNOLOGY

Design a Solution for Backup and Recovery

Learning Objectives

By the end of this lesson, you will be able to:

- Identify the best practices for reliability to enhance system resilience and ensure continuous accessibility
- Gain insights into Microsoft Azure backup to offer simple and reliable backup and protection for critical data
- Analyze and understand Azure to Azure Site Recovery to replicate Azure VMs to any other Azure location
- List the components of Azure disaster recovery to ensure comprehensive protection and rapid recovery
- Summarize the working of failover process architecture to design and implement robust disaster recovery solutions

TECHNOLOGY

Recommend a Recovery Solution for Azure Workloads

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Introduction

Reliability consists of two principles: **resiliency** and **availability.**

The goal of resiliency is to prevent failures and, if they persist, restore the application to full functionality.

Availability aims to ensure consistent access to the application or workload.

Architectural Best Practices for Reliability

- Define requirements
- Use architectural best practices
- Test with simulations and forced failovers
- Deploy the application consistently
- Monitor application health
- Respond to failures and disasters

Define Requirements

Define requirements using the given criteria:

1. Identify workloads and usage
2. Plan for usage patterns
3. Establish availability metrics (MTTR/MTBF)
4. Establish recovery metrics (RTO/RPO)
5. Determine workload availability targets
6. Understand service-level agreements

Recovery Time Objective (RTO)

It is the maximum acceptable time for which an application can be unavailable after an incident.

- If the RTO is 90 minutes, the application must be restored to a running state within 90 minutes from the start of a disaster.
- If an application has a very low RTO, consider a second deployment running in standby mode.
- If SQL Server and its databases can be brought online in five minutes but application servers take 20 minutes to do the same, the overall RTO would be 20 minutes, not five.

Recovery Point Objective (RPO)

It is the maximum duration of data loss that is acceptable during a disaster.

Example:

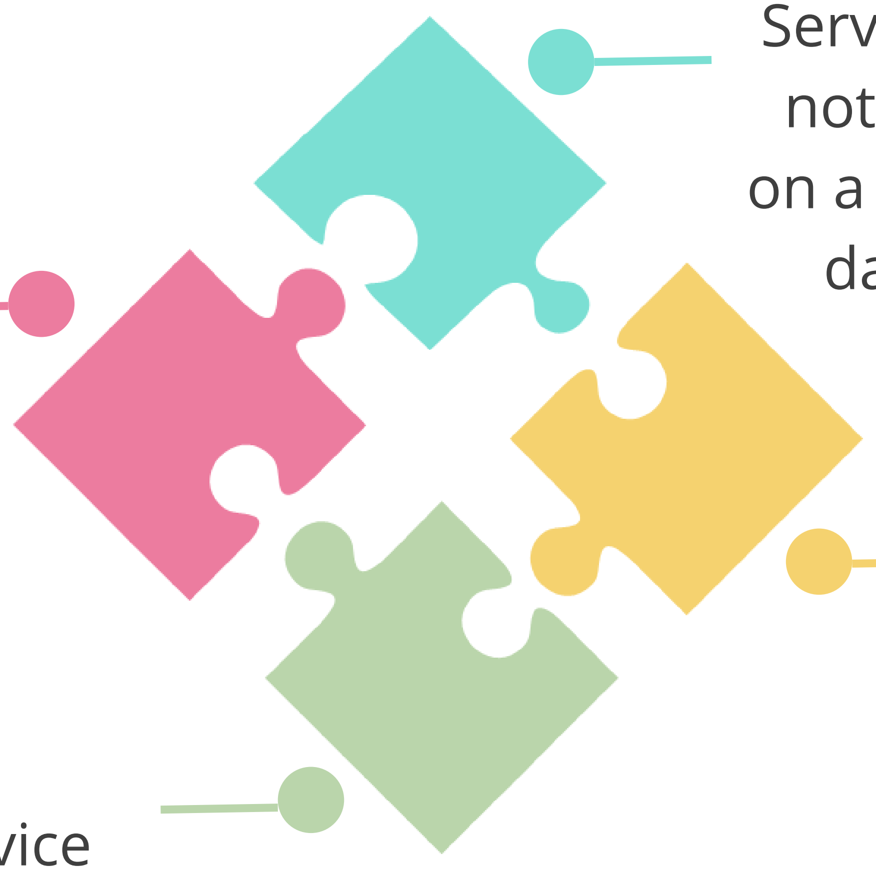
Data stored in a single database with no replication to other databases and hourly backups could result in the loss of an hour's worth of data.

Use Architectural Best Practices

- Perform a Failure Mode Analysis (FMA)
- Create a redundancy plan
- Design for scalability
- Plan for subscription and service requirements
- Use load-balancing to distribute requests
- Build availability requirements into the design
- Manage the data

Azure Service Dependencies

The following Azure services are dependent on one another:



Services that are not dependent on a single logical data center

Customer-specific regions

Non-regional services deployed to two or more regions

Azure service dependency documentation

Test with Simulations and Forced Failovers

The steps involved in simulating a common failure scenario are as follows:

- Simulate common failure scenarios
- Identify failures that occur only under load
- Run disaster recovery drills
- Perform failover and failback testing
- Run simulation tests
- Test health probes
- Test monitoring systems
- Include third-party services in test scenarios

Deploy the Application Consistently

The stages in deployment are as follows:

- Automate the application deployment process
- Design the release process to maximize availability
- Have a rollback plan for deployment
- Log and audit deployments
- Document the application release process

Monitor Application Health

The following factors are often used to take care of the application's health:

- Measure remote call statistics and share the data with the application team
- Track transient exceptions and retries over an appropriate time frame
- Implement health probes, check functions, and monitor third-party services
- Train multiple operators to monitor the application and perform manual recovery steps

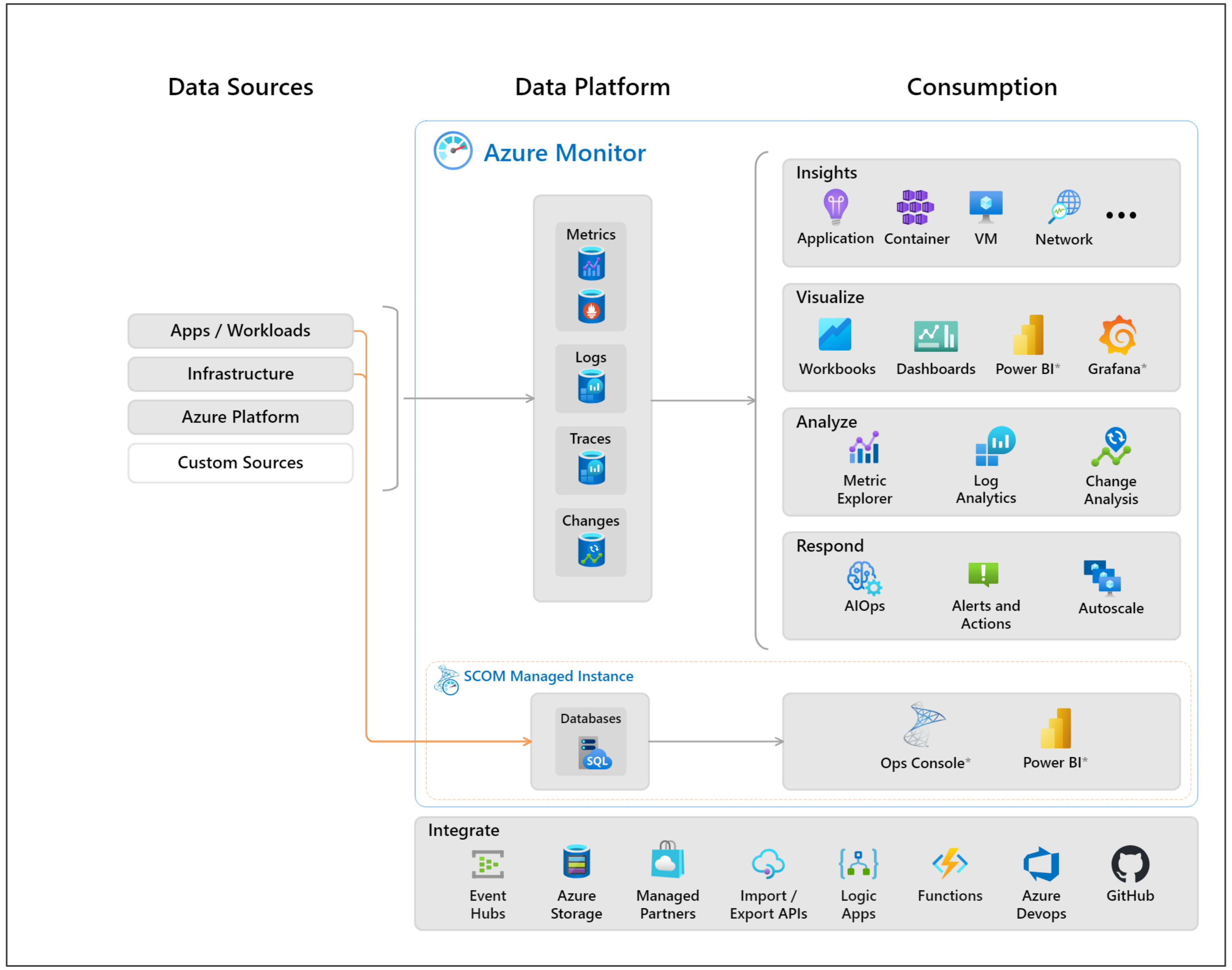
Monitor Application Health

The following factors are often used to take care of the application's health:

- Check long-running workflows
- Maintain application logs
- Set up an early warning system
- Operate within Azure subscription limits

Monitor Application Health

The following is the Azure Monitor framework for comprehensive application health monitoring:



Respond to Failures and Disasters

In case the application fails, the following operations are carried out:

- Plan for Azure support interactions
- Document and test your disaster recovery plan
- Failover manually when required
- Prepare for application failure

Respond to Failures and Disasters

In case the application fails, the following operations are carried out:

- Recover from data corruption
- Recover from a network outage
- Recover from a dependent service failure
- Recover from a region-wide service disruption

TECHNOLOGY

Recommend a Solution for Azure Backup Management

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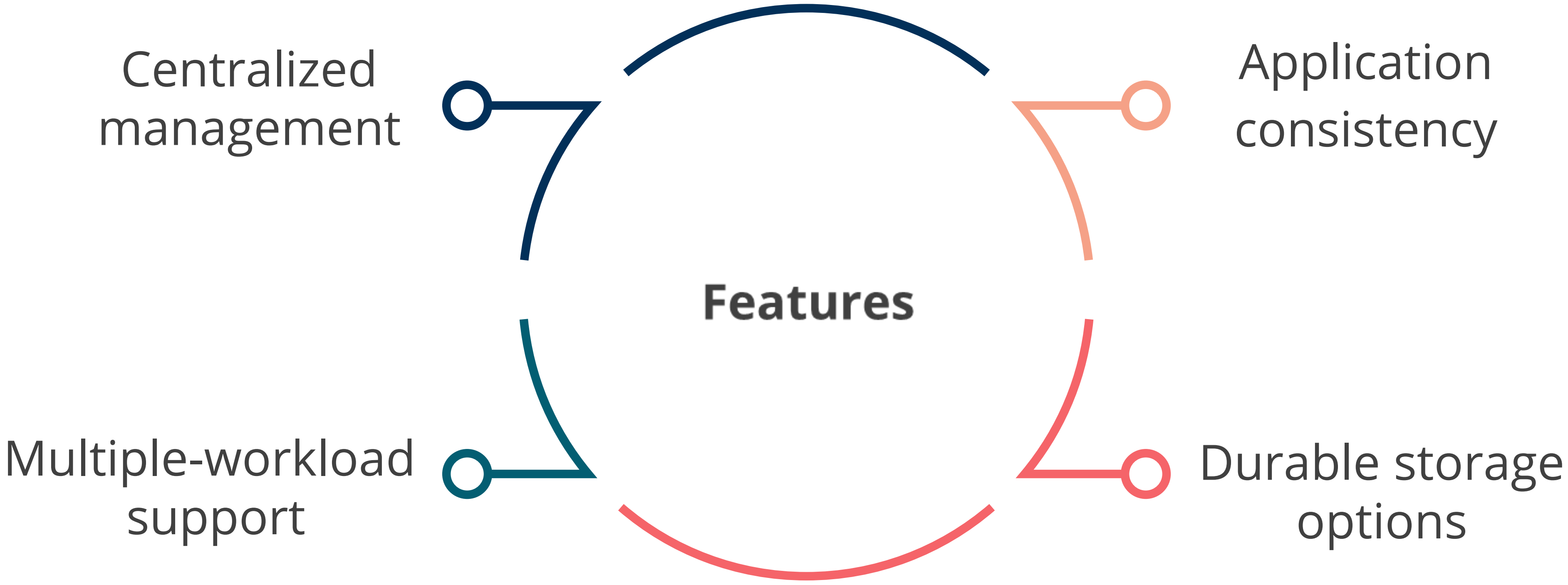
Microsoft Azure Backup

It offers simple and reliable backup and protection for critical data in an easily recoverable way from any location.



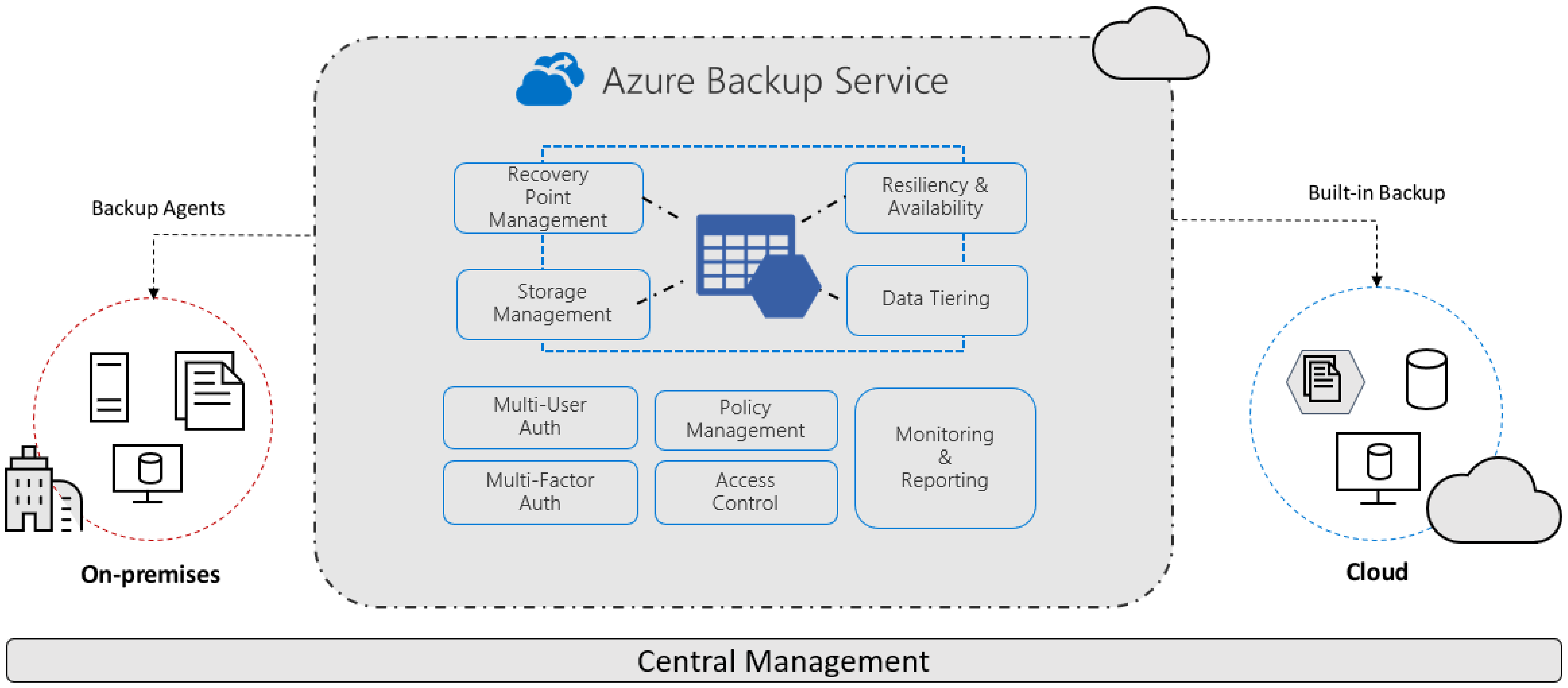
Microsoft Azure Backup: Benefits

- Offload on-premises backup
- Back up Azure IaaS VMs
- Scale easily
- Get unlimited data transfer

- Keep data secure
- Get app-consistent backups
- Multiple storage options
- Zone-redundancy

Azure VM Backup Architecture

The following figure illustrates the architecture of the Azure Backup Service:



Source: https://learn.microsoft.com/en-us/azure/backup/backup-overview

Azure Backup: Key Features

- Simple configuration and management
- Block-level incremental backups
- Data integrity verified in the cloud
- Configurable retention policies

A simple and familiar user interface to configure and monitor backups from Windows Server and System Center Data Protection Manager

Azure Backup: Key Features

- Simple configuration and management
- Block-level incremental backups
- Data integrity verified in the cloud
- Configurable retention policies

Automatic incremental backup track file and block-level changes, only transferring the changed blocks, hence reducing the storage and bandwidth utilization

Azure Backup: Key Features

- Simple configuration and management
- Block-level incremental backups
- Data integrity verified in the cloud
- Configurable retention policies

Backed-up data is also automatically checked for integrity once the backup is complete. As a result, any corruptions due to data transfer are automatically identified, and repair is attempted in the next backup.

Azure Backup: Key Features

- Simple configuration and management
- Block-level incremental backups
- Data integrity verified in the cloud
- **Configurable retention policies**

Retention policies are used to control how long a backup will be saved in Azure. This helps to meet business policies and manage backup costs.

Azure IaaS Backup Components

Component	Benefits	Limits	What is protected?	Where are backups stored?
Azure Backup Server (can be deployed in Azure and on-premises)	• App aware snapshots (VSS) • Full flexibility for when to take backups • Recovery granularity (all) • Can use Recovery Services vault • Linux support on Hyper-V and VMware VMs • Backup and restore VMware VMs • Does not require a System Center license	• Cannot backup Oracle workload • Always requires live Azure subscription • No support for tape backup	• Files • Folders • Volumes • VMs • Applications • Workloads • System State	• Recovery Services vault • Locally attached disk
Azure IaaS VM Backup	• Native backups for Windows/Linux • No specific agent installation required • Fabric-level backup with no backup infrastructure needed	• Backup VMs once a day • Restore VMs only at disk level • Cannot back up on-premises	• VMs • All disks (using PowerShell)	Recovery Services vault

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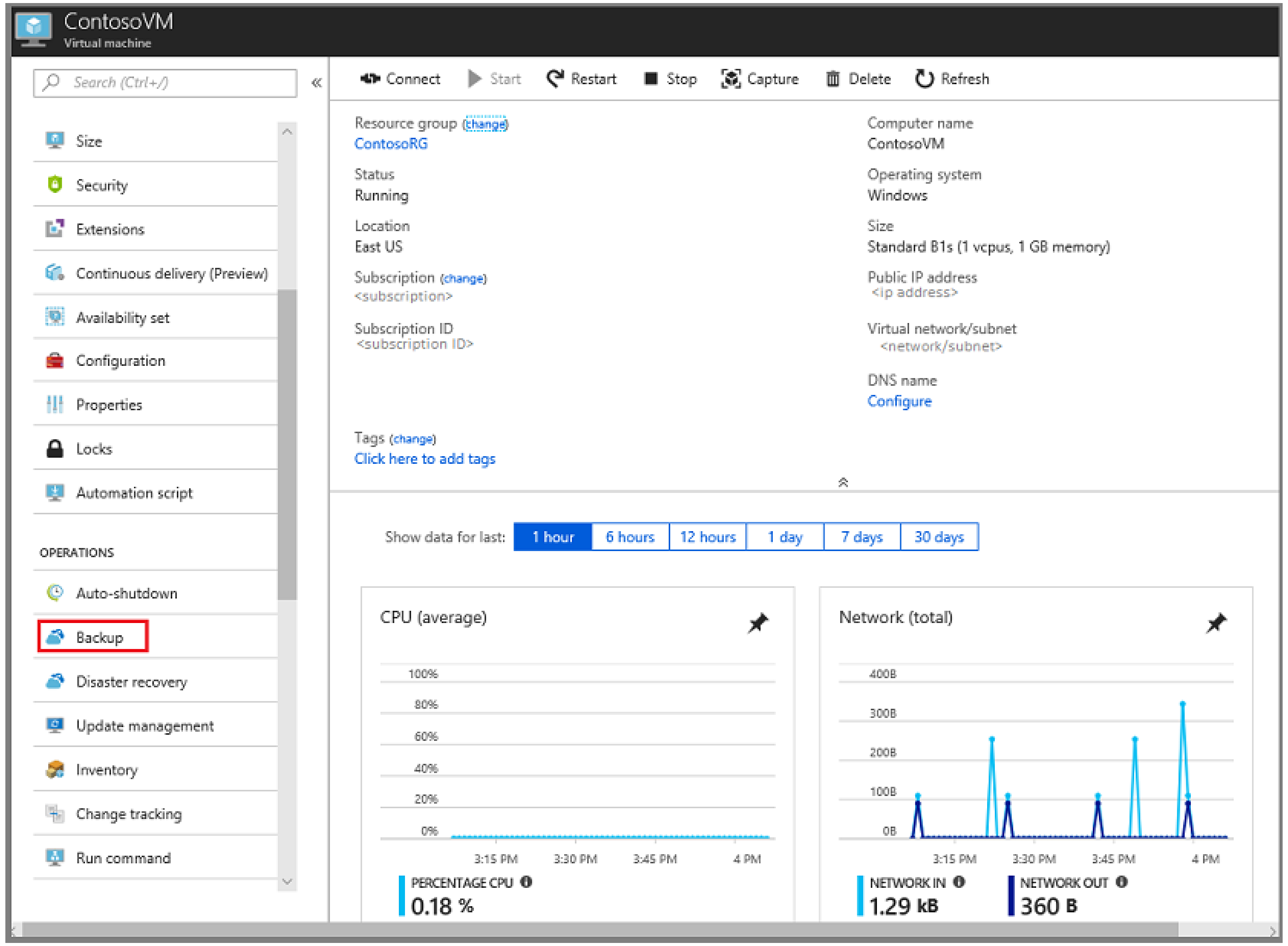
Recovery Services Vault

It is a storage entity in Azure that stores backup data.



Azure Blob Backup

It is a managed, local data protection solution that will protect user block blobs from various data loss scenarios like corruptions, blob deletions, and accidental storage account deletion.

Facilitates point-in-time recovery with local storage

Optimizes storage centralization in Azure Backup vault

Enhances simplicity, security, and cost-effectiveness

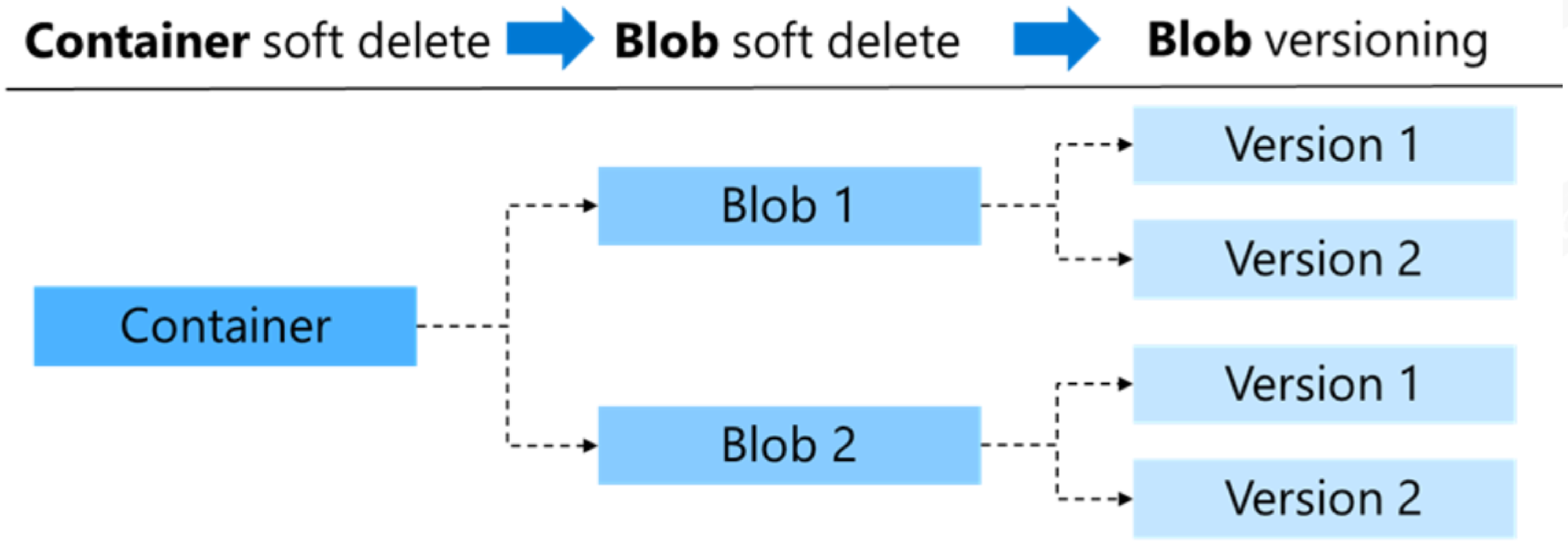
Data Protection in Blob Service

The properties listed in the data protection tab of the blob service within the user's storage account are:

- Point-in-time restore
- Versioning for blobs and blob change feed
- Soft delete
- Delete lock

Design for Azure Blob Backup

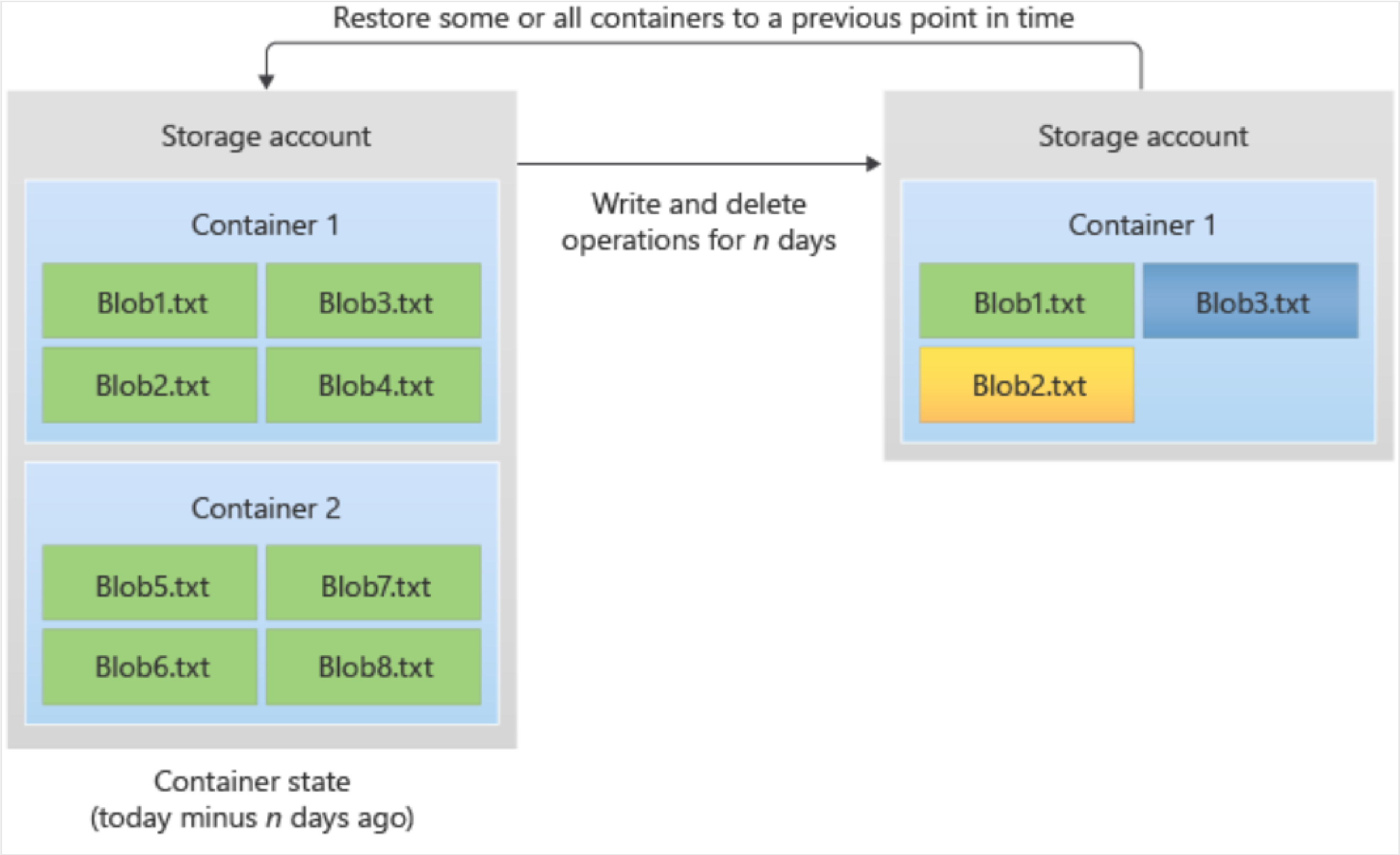
Soft delete protects an individual blob, snapshot, container, or version from accidental deletes or overwrites.



Advantages of Soft Delete and Versioning

Container soft delete	Restores a container and its contents at the time of deletion Retention period: Between 1 and 365 days (for deleted containers)
Blob soft delete	Restores a blob, snapshot, or version that has been deleted Retention period: Between 1 and 365 days (for deleted blobs)
Blob versioning	Restores an earlier version of a blob and recovers the data if it is incorrectly modified or deleted

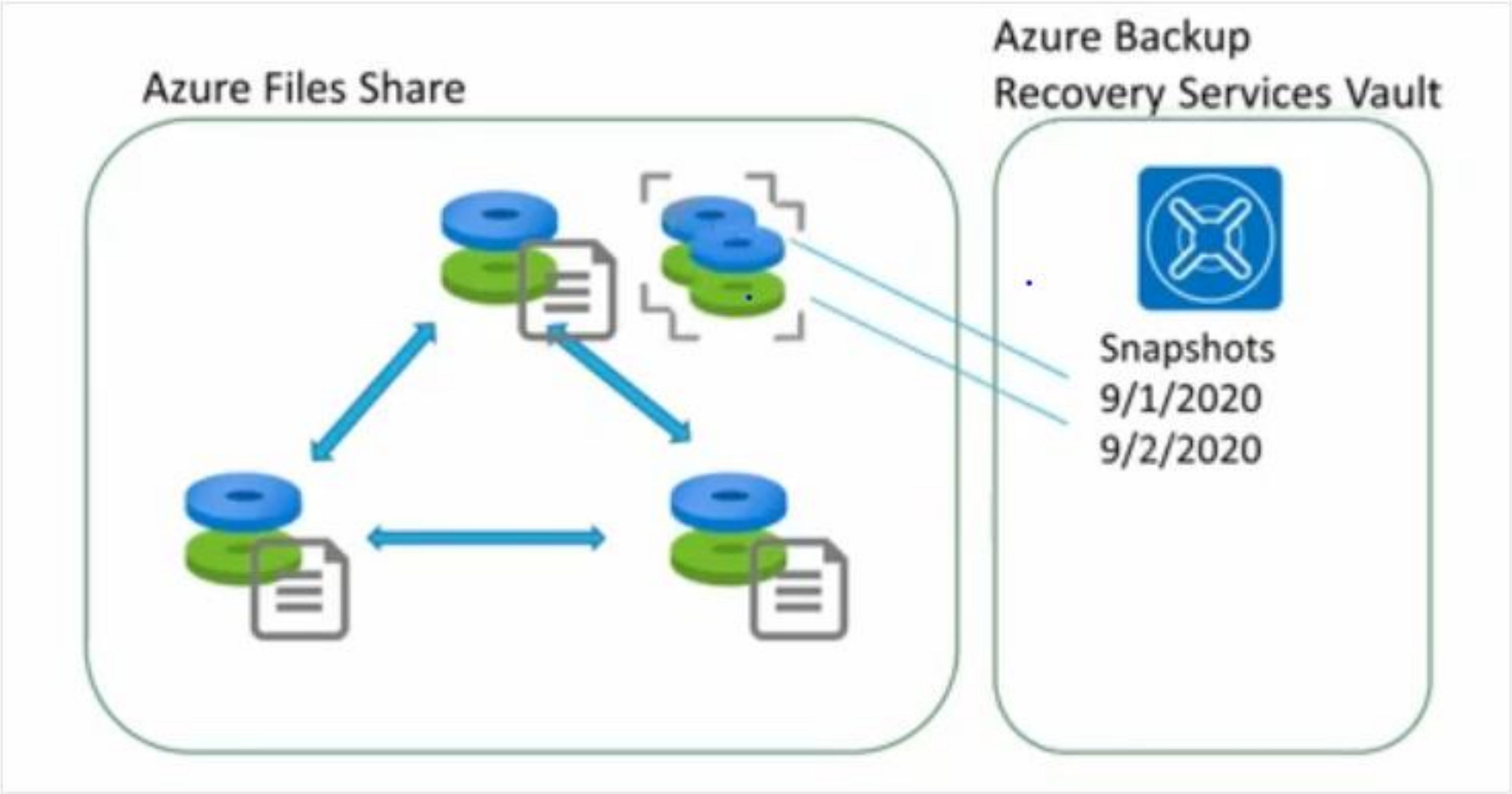
Point-in-Time Restore for Block Blobs



The prior state of one or more containers or blob ranges is restored. As a result, write and delete actions that occurred during the retention period are reversed.

Azure Files Backup

Azure files can capture share snapshots of file shares.



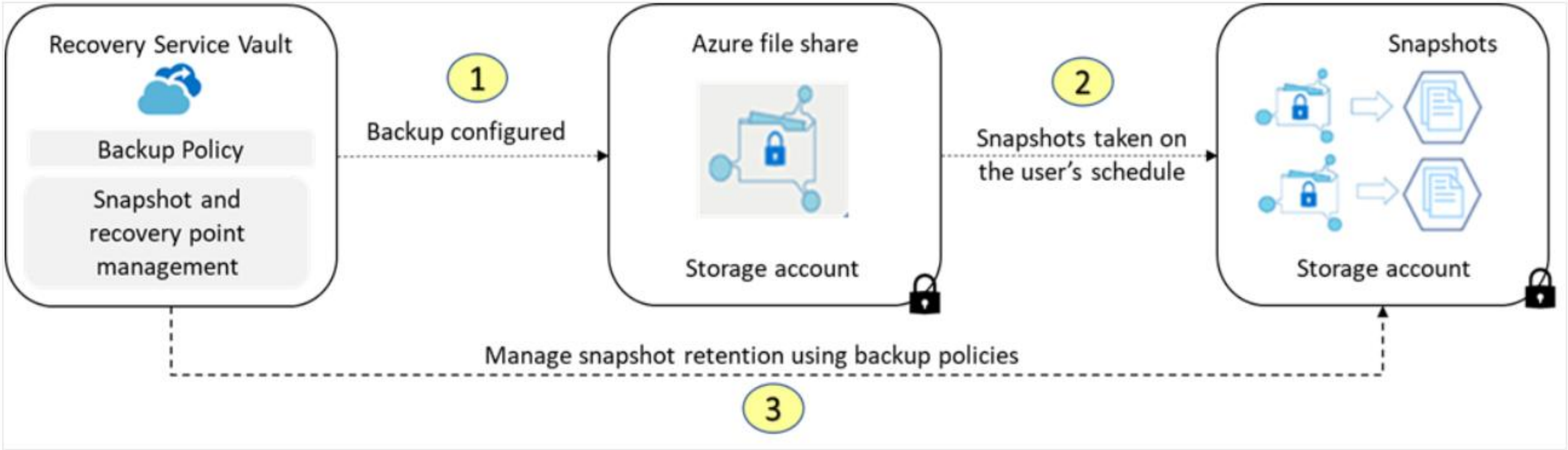
Share snapshots provide an added layer of security, lowering the danger of data corruption or deletion by accident.

Benefits of Snapshots

- Capture the state of the share at a specific moment with share snapshots
- Automate snapshot creation using Azure Backup and backup policies
- Create snapshots manually through the Azure portal, REST API, client libraries, Azure CLI, or PowerShell
- Store snapshots at the root level, affecting all folders and files; retrieve on a file-by-file basis
- Read, copy, or delete share snapshots after creation; modification is prohibited

Automating File Share Backups

The metadata of the backup is kept in the recovery services vault by Azure Backup, but no data is sent. This entails a quick backup solution with backup and built-in reporting.



Azure SQL Backup: Types

Type	Description
Full backups	Everything in the database and the transaction logs is backed up. Backup: every week
Differential backups	Everything that changed since the last full backup is backed up. Backup: every 12 - 24 hours
Transactional backups	The data is restored up to a specific time, which includes the moment before it was mistakenly deleted. Backup: every 5 to 10 minutes

Backup Use Cases

The following are some backup use cases and their respective explanations and limitations:

Uses cases	Explanation	Limitations
Restore an existing database to a point in time in the past	To avoid overwriting the original database, this procedure generates a new database on the same server as the original database, but with a different name.	New database must be on the same server; cannot overwrite the original
Restore a deleted database to the time of deletion	The deleted database can only be restored on the same server or managed instance where it was created in the first place.	Restoration is limited to the original server or managed instance

Backup Use Cases

The following are some backup use cases and their respective explanations and limitations:

Uses cases	Explanation	Limitations
Restore a database to another geographic region	During a geographic disaster, if the user can't access the primary region's database or backups, they can quickly recover using geo-restore, creating a new database on any Azure region with an existing server or managed instance.	Recovery is dependent on the availability of an existing server or managed instance in the alternate region
Restore a database from a specific long-term backup	Users can restore an older version of the database if the database has been set up with a long-term retention policy.	Requires a configured long-term retention policy to restore older versions

Long-Term Backup Retention Policies

Recovery Time Objective (RTO) is the maximum acceptable time that an application can be unavailable after an incident.

- Backups of Azure SQL Database can be stored in read-access geo-redundant storage (RA-GRS) blobs for up to ten years.
- To access a backup in LTR, a user may use the Azure interface or PowerShell to restore it as a new database.

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Design a Solution for Data Archiving and Retention

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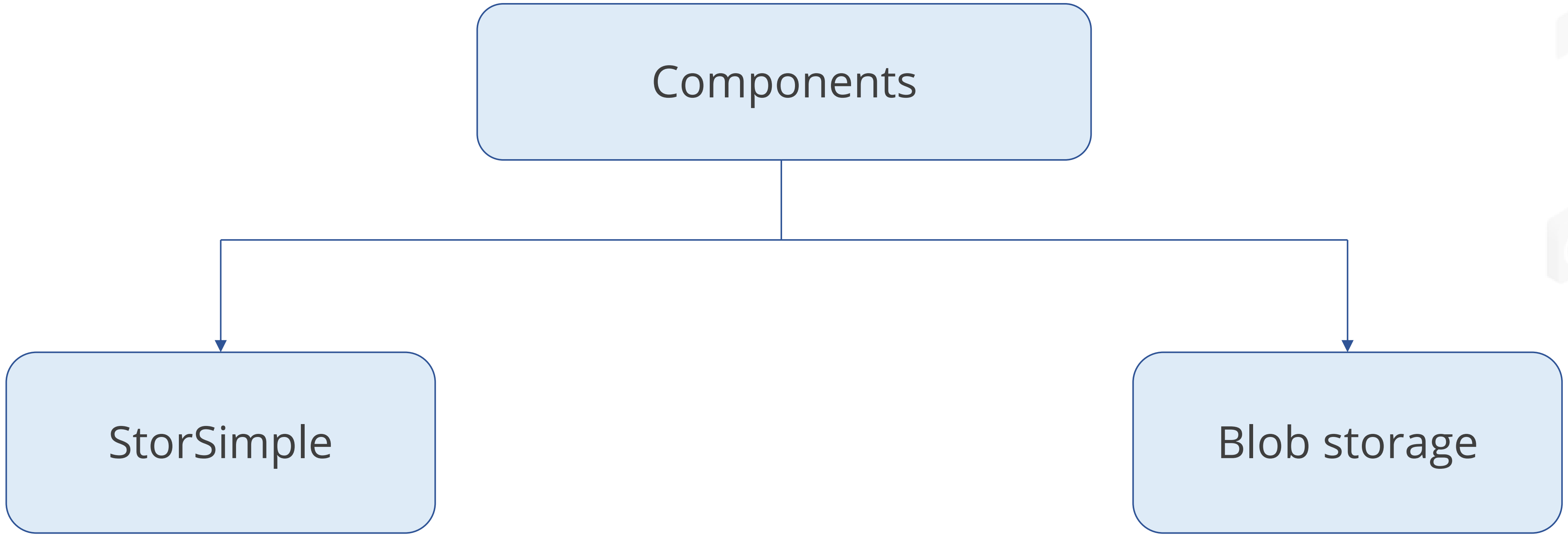
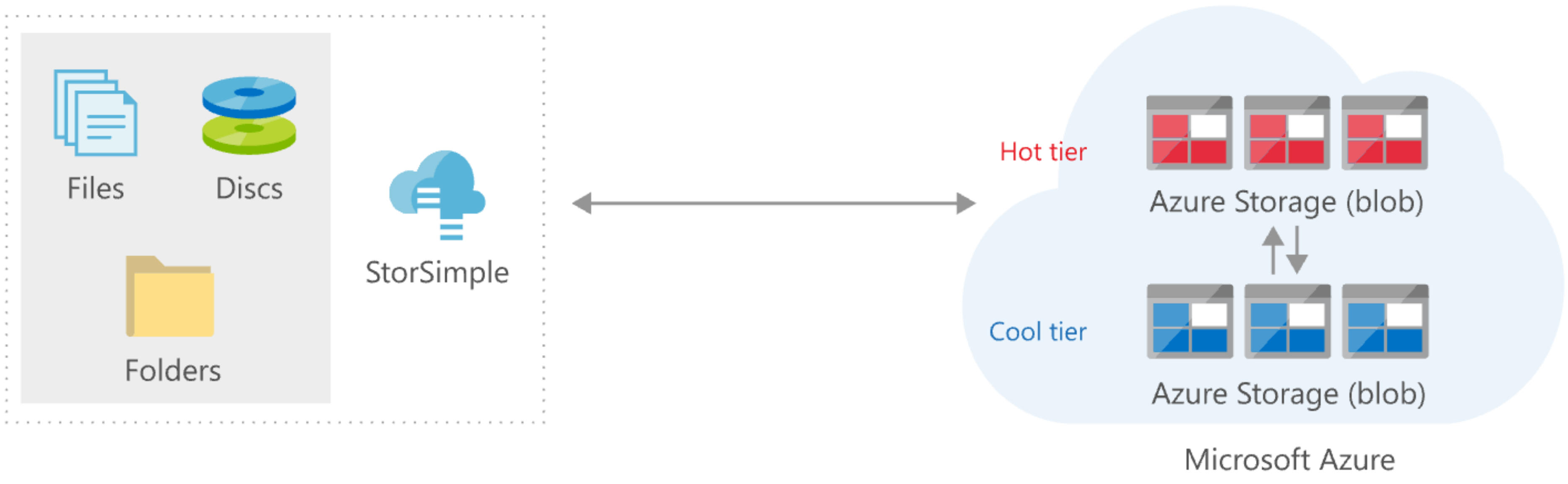
Archive On-Premises Data to Cloud

Azure Blob storage can be used to archive on-premises data.



Archive On-Premises Data to Cloud

The Azure StorSimple appliance running on-premises can tier data to Azure Blob storage (both hot and cool tier).

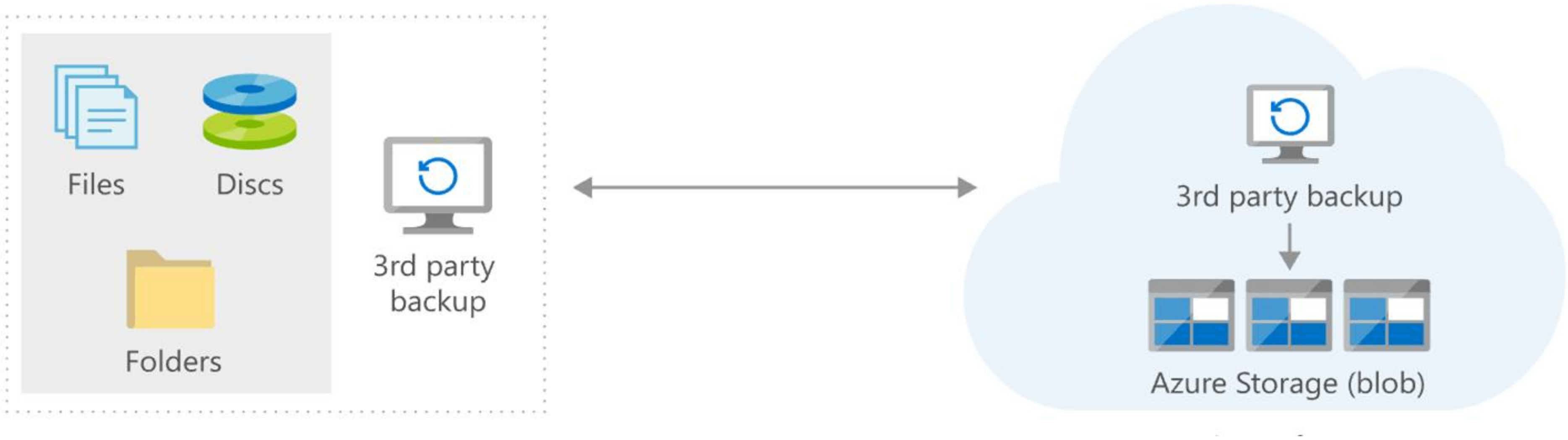


Azure StorSimple

StorSimple can be used to archive data from on-premises to Azure.

Archive On-Premises Data to Cloud

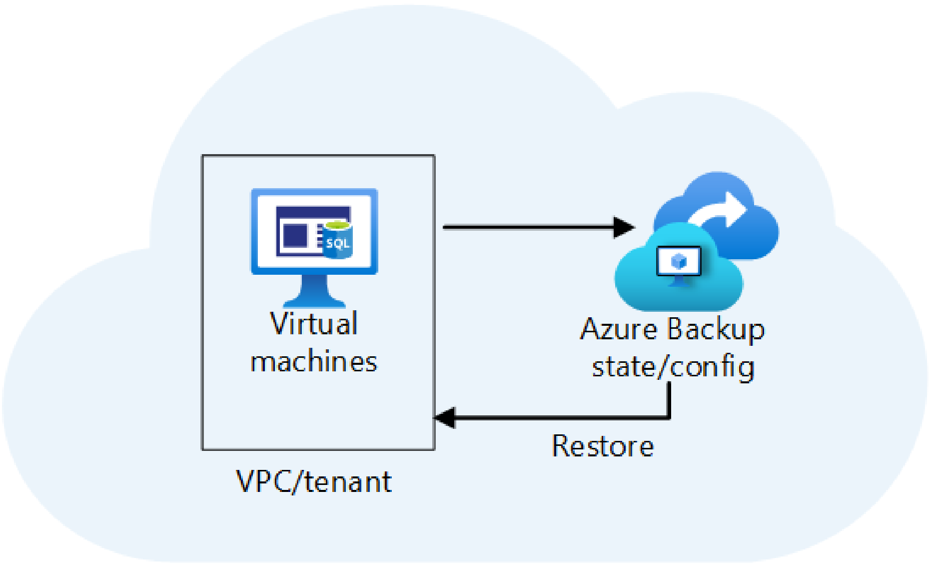
Blob storage is a cool or archive tier on Azure Blob storage, which is used to back up data that is less frequently accessed, while a hot tier is used to store data that is frequently accessed.



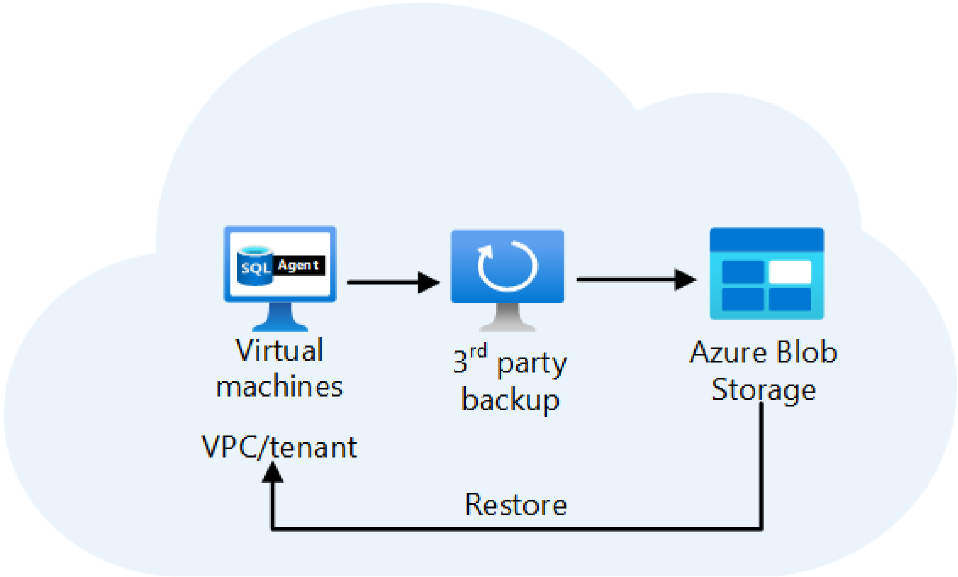
Azure Blob Storage

Backup On-Premises Applications and Data to the Cloud

Back up data and applications from an on-premises system to Azure using Azure Backup or a partner solution



Option 1 – Use native Azure Backup



Option 2 – Use 3rd party backup apps

Microsoft Azure

Source: https://learn.microsoft.com/en-us/azure/architecture/solution-ideas/articles/backup-archive-on-premises-applications

Backup On-Premises Applications and Data to the Cloud

The components are:

Azure Backup Server	Manages the configuration of restore procedures and orchestrates machine backups
Azure Backup Service	Runs on the cloud and holds the recovery points, enforces policies, and manages data and application protection
Blob Storage	Backs up data and applications for partner solutions such as Commvault

TECHNOLOGY

Design an Azure Site Recovery Solution

Azure to Azure Site Recovery (ASR)

Replicates Azure VMs to any other Azure location

Replicates:

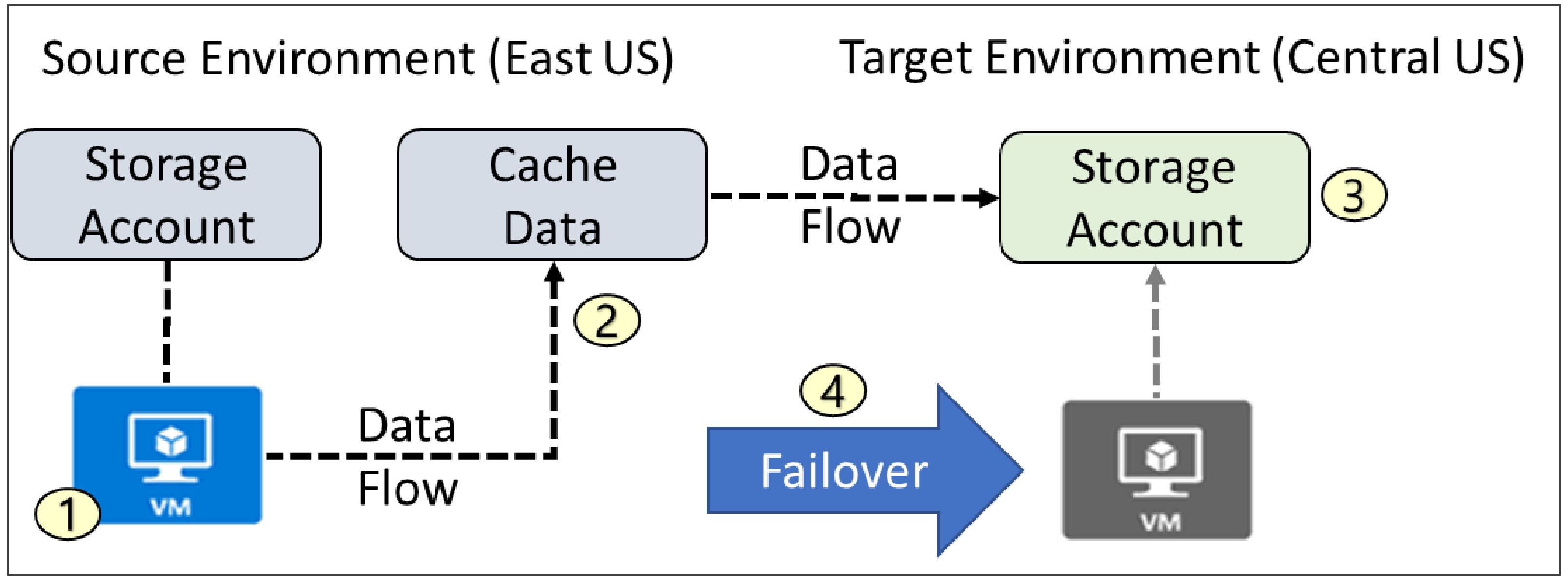
- VMs
- Virtual Networks
- Availability Sets
- Storage accounts
- Optionally – User can specify own VNets, storage accounts and availability sets

Replicates HTTPS channel on 443:

Generates Azure egress cost for outbound traffic from the primary region

It automatically deploys ASR Mobility Service extension to protected machines.

Steps: ASR Architecture



- VM is registered with Azure Site Recovery
- Data is continuously replicated to cache
- Cache is replicated to the target storage account
- During failover, the virtual machine is added to the target environment

Azure to ASR: Network Traffic

ASR traffic is only outbound from protected VMs.

Access to ASR URLs:

*.blob.core.windows.net	For writing data to the cache storage account
login.microsoftonline.com	For authorization and authentication to the ASR URLs
*.hypervrecoverymanager.windowsazure.com	For ASR service communication channel
*.servicebus.windows.net	For monitoring and diagnostics

Azure to ASR: Network Traffic

ASR traffic is only outbound from protected VMs.

Outbound security rules for NSGs:

Site Recovery service and monitoring IP addresses (location specific):

Location	Site Recovery service IPs	Site Recovery monitoring IP
Central US	40.69.144.231	52.165.34.144

Snapshots and Recovery Points

Recovery points are created from snapshots.

Site Recovery snapshots:

Crash-consistent

Data that was on the disk when the snapshot was taken

App-consistent

All the information in a crash-consistent snapshot, data in memory, and transactions in progress

Creating Azure Site Recovery

Duration: 5 Min.

ASSISTED PRACTICE

Problem Statement:

As an Azure architect, you have been asked to assist your company with an Azure backup and recovery solution. Your data should be safe and recoverable with this Azure backup solution.

Outcome:

You will be able to successfully implement an Azure Site Recovery solution that ensures all data is securely backed up and recoverable, leveraging Azure's robust and scalable architecture.

Note: Refer to the demo document for detailed steps:
01_Creating_Azure_Site_Recovery

Assisted Practice: Guidelines

Steps to create Azure site recovery are:

1. Create Azure site recovery

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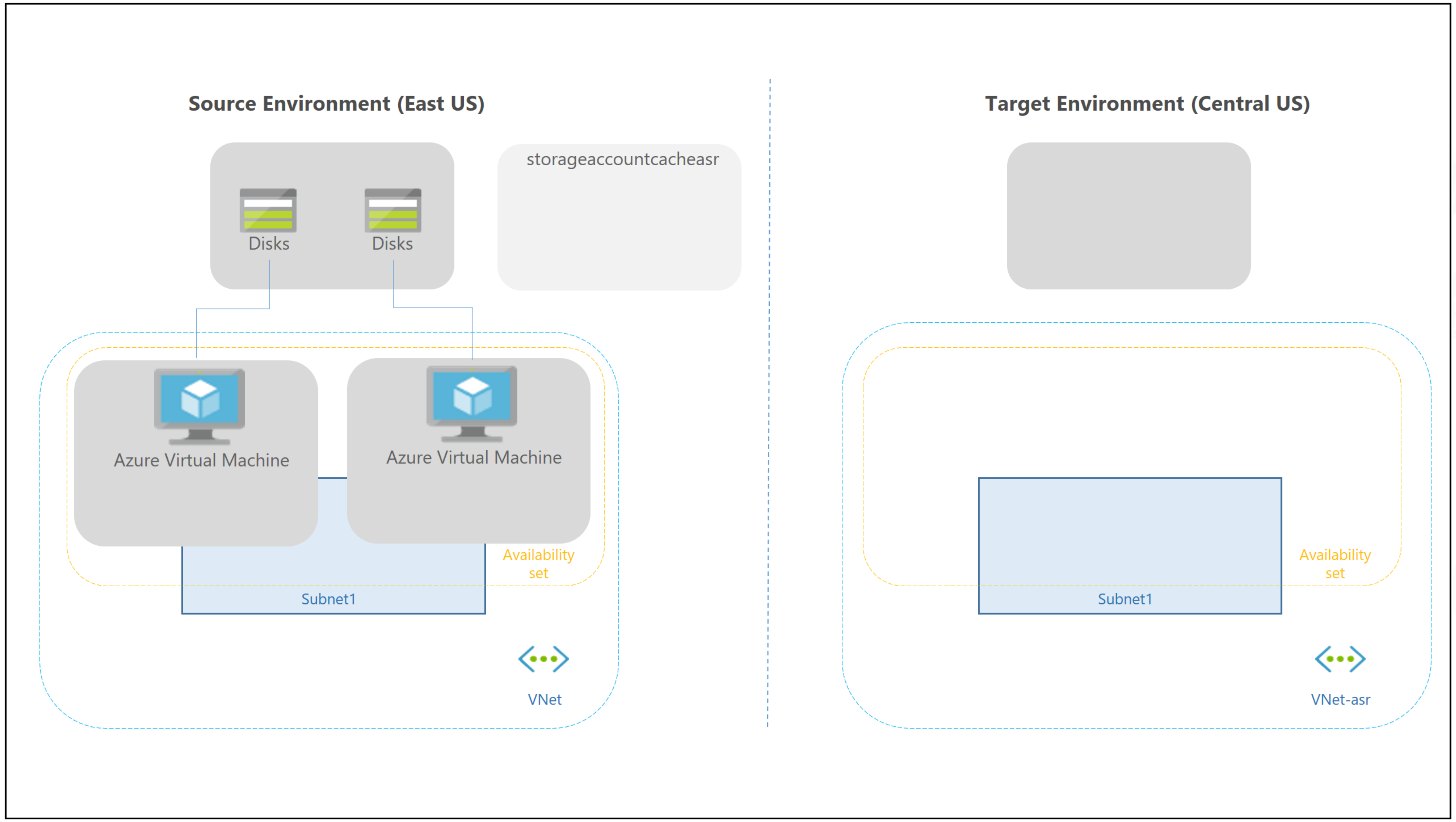
Recommend a Solution for Recovery in Different Regions

Azure to Azure Disaster Recovery Architecture

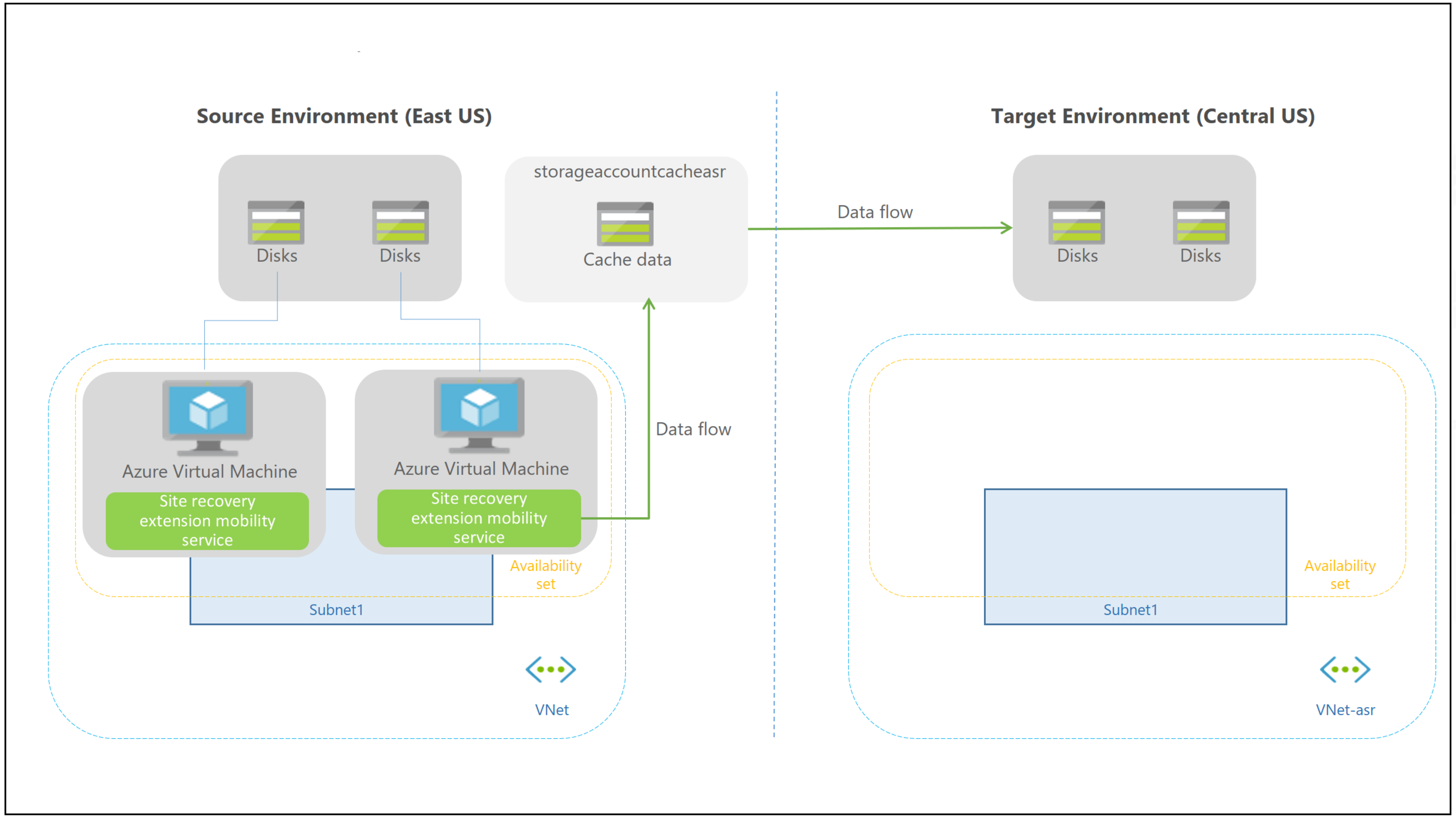


Source: https://learn.microsoft.com/en-us/azure/site-recovery/azure-to-azure-architecture

Azure Disaster Recovery: Architectural Components

Components	Requirements
VMs in the source region	One or more Azure VMs must be in a supported source region.
Source VM storage	Non-managed disks can be distributed through storage accounts, whereas managed disks can be used in Azure VMs.
Source VM networks	In the source field, VMs can be found in one or more subnets of a virtual network.
Cache storage account	Using a cache means that production applications running on a VM have minimal effects.
Target resources	Target resources are used during replication and failover.

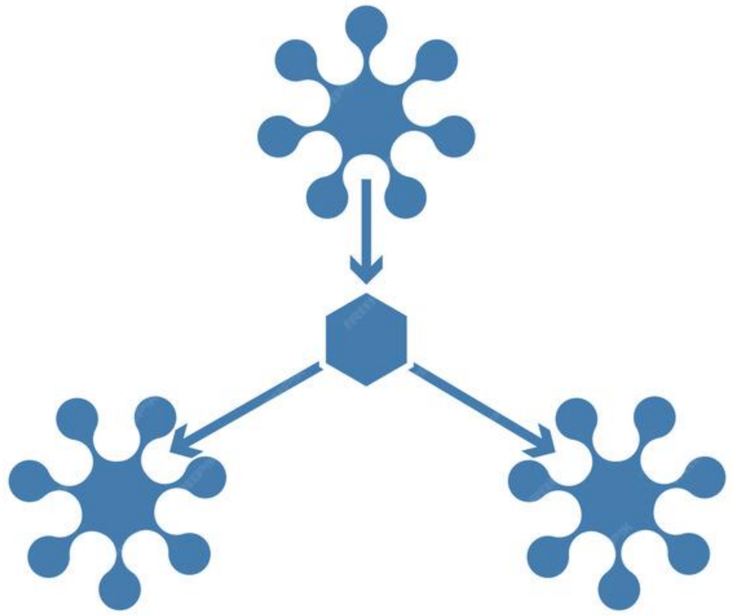
Replication Process Architecture



Source: https://learn.microsoft.com/en-us/azure/site-recovery/azure-to-azure-architecture

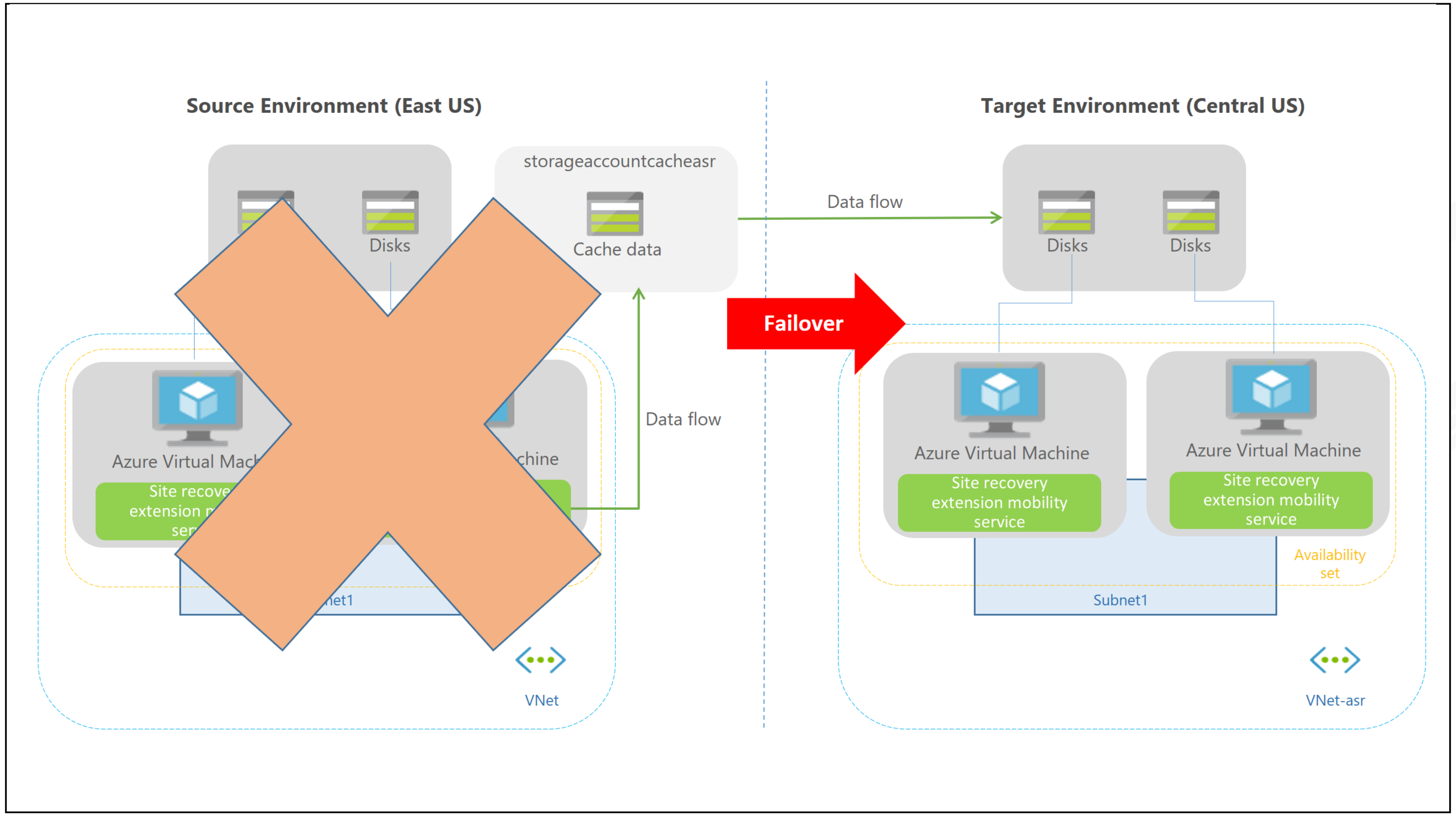
Replication Process

When replication is enabled for an Azure VM:



- The site recovery mobility service extension is installed on the VM.
- The extension registers the VM with site recovery.
- Continuous replication begins for the VM.
- Disk writes are transferred to the cache storage.
- Data is processed and sent to a target storage account or a replicated disk.
- Crash-consistent recovery points are generated every five minutes after data processing.

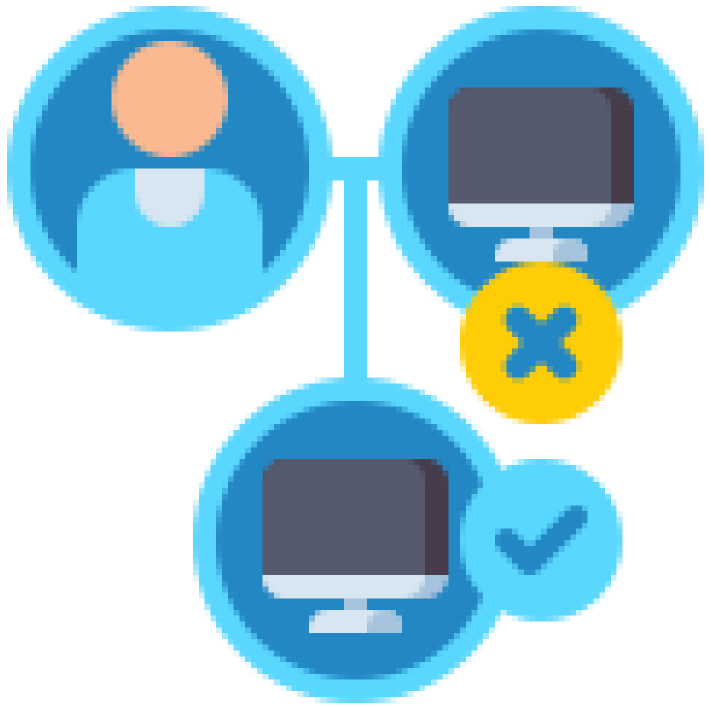
Failover Process Architecture



Source: https://learn.microsoft.com/en-us/azure/site-recovery/azure-to-azure-architecture

Failover Process

When a user initiates a failover, the VMs are created in the target resource group, virtual network, subnet, and availability set.



The user can use any recovery point during a failover.

Key Takeaways

- Defining the requirements, deploying the application regularly, and monitoring the health are all important facets in sustaining reliability.
- Recovery time objective (RTO) is the maximum acceptable time for which an application can be unavailable after an incident.
- The architectural components of disaster recovery are VMs in the source region, their storage and networks, cache storage account, and target resources.
- The two important components required to back up cloud applications and data to the cloud are Azure Backup Service and Blob Storage.

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Knowledge Check

Knowledge Check 1	Which of these steps is involved in simulating a common failure scenario?

A. Run disaster recovery drills

B. Test health probes

C. Include third-party services in test scenarios

D. All of these

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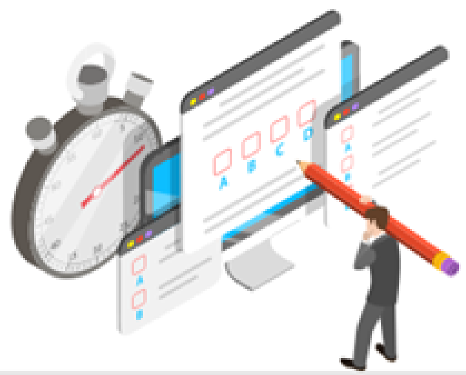


The correct answer is **D**

The steps involved in simulating common failure scenarios include running disaster recovery drills, testing health probes, and including third-party services in test scenarios.

Knowledge Check

2

Which of the following is NOT true for the replication process?

A. Data is processed and sent to the target storage account.

B. Disk writes are transferred to the disk storage.

C. The site recovery mobility service extension is installed on the VM.

D. None of the above

Knowledge Check 2

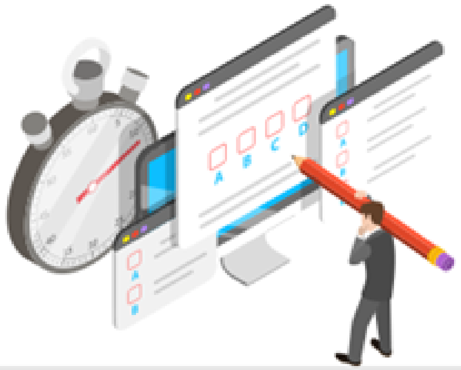
Which of the following is NOT true for the replication process?

A. Data is processed and sent to the target storage account.

B. Disk writes are transferred to the disk storage.

C. The site recovery mobility service extension is installed on the VM.

D. None of the above



The correct answer is **B**

During replication, disk writes are transferred to the cache storage.

Knowledge Check 3	**Which of the following is NOT an architectural component of disaster recovery?**

A. Source VM storage

B. Cache storage account

C. Target resources

D. Azure VMs in a non-supported source region

Knowledge Check 3

Which of the following is NOT an architectural component of disaster recovery?

A. Source VM storage

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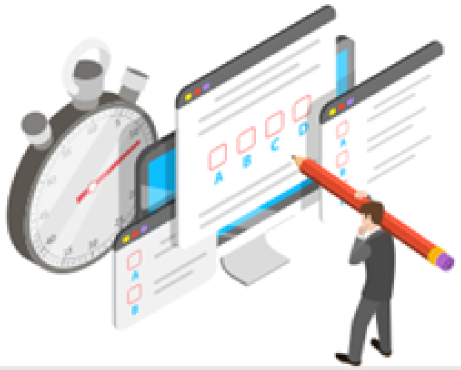
D. Azure VMs in a non-supported source region



The correct answer is **D**

Azure VMs in a non-supported source region is not an architectural component of disaster recovery.

Knowledge Check 4	Which of these is required to back up cloud applications and data to the cloud?

A. Azure Backup Service

B. Azure Blob Storage

C. Both A and B

D. Neither A nor B

Knowledge Check 4

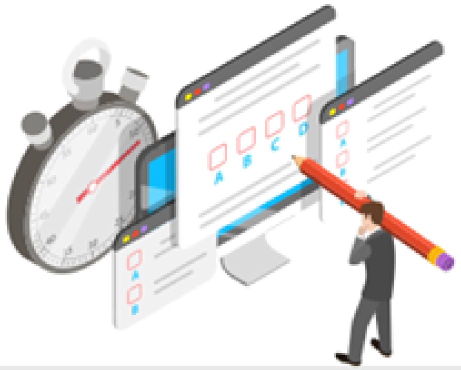
Which of these is required to back up cloud applications and data to the cloud?

A. Azure Backup Service

B. Azure Blob Storage

C. Both A and B

D. Neither A nor B



The correct answer is **C**

The two important components required to backup cloud applications and data to the cloud are Azure Backup Service and Blob Storage.

Azure Site Recovery between Azure Regions

Duration: 45 min.

LESSON-END PROJECT

Project agenda: To implement Azure Site Recovery between Azure regions

Description: You need to implement Azure Site Recovery to facilitate the migration and protection of Azure VMs between regions. Begin by deploying an Azure VM to be migrated using an Azure resource manager template and creating an Azure Recovery Services vault. Afterward, configure Azure VM replication and review the Azure VM replication settings.

Perform the following:

1. Deploy an Azure VM
2. Create an Azure Recovery Services vault
3. Configure Azure VM replication
4. Review Azure VM replication settings

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Thank You